Matthew Justin McDowell

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Electrical Engineer with grid operations experience at ERCOT — specializing in energy storage resource monitoring, ancillary services, and EMS tooling. Targeting BESS integration and grid storage roles in the Pacific Northwest.

Core Competencies

- Energy Storage Resource Operations
- EMS / Load Frequency Control (LFC)
- Python | MATLAB | Pandas | PyTorch
- PSSE | PowerWorld | ArcGIS | Cadence
- Battery Dispatch & Ancillary Services
- NERC Standards & ERCOT Protocols
- Machine Learning & Statistical Analysis
- Power System Fault Analysis & Modeling

Experience

Operations Engineer **ERCOT** *Austin, TX 09/2024 – Present*

- Monitor and support real-time operations of the ERCOT grid, with direct responsibility for Energy Storage Resource (ESR) performance tracking, dispatch behavior, and compliance with NERC Standards and ERCOT Protocols.
- Tune and maintain Energy Management System components — Load Frequency Control (LFC), Resource Limit Calculator (RLC), and Generation to be Dispatched (GBD) — to ensure reliable frequency regulation across the grid.
- Develop and apply statistical, probabilistic, and machine learning methods for Ancillary Service quantification, directly informing procurement decisions for Regulation, Responsive Reserve, and Non-Spin services.
- Build Python-based tools and automated workflows that enhance real-time situational awareness, reduce manual analysis burden, and improve grid reliability monitoring.
- Analyze frequency control performance and IRR variability to identify system risks; collaborate across departments on model updates, market rule changes, and reliability studies.

Electrical Engineer Intern **EDP Renewables** *Houston, TX 05/2023 – 08/2023*

- Built Python automation from scratch for the generator interconnection application process, eliminating tens of hours of manual work per application previously required to produce one-line diagrams, PSSE models, and Excel deliverables.
- Designed and deployed UiPath RPA automations for the Transmission and Interconnection department, reducing repetitive manual task load across the team.
- Mapped generation interconnection upgrade requirements for prospective renewable and storage project sites using ArcGIS and Python API — directly supporting BESS siting decisions.

IT Support Engineer Intern **Amazon** *Richmond, VA 05/2022 – 08/2022*

- Deployed atmospheric monitoring for the main distribution frame; resolved infrastructure issues across hardware, software, and network systems; developed new support procedures adopted company-wide.

Student IT Technician **Texas A&M IT** *College Station, TX 08/2021 – 12/2023*

Education

B.S. Electrical Engineering — 3.9 GPA **Texas A&M University** *College Station, TX 08/2019 – 12/2023*

Minors: Computer Science & Mathematics

Relevant Coursework: Power System Fault Analysis, Power System Operation & Control, Energy Conversion, Properties of Materials, Data Structures & Algorithms, Machine Learning, Digital Circuit Design

Projects & Technical Work

- **Battery Dispatch Optimization Model (In Progress):** Python simulation modeling BESS state-of-charge, round-trip efficiency, and ancillary service revenue stack for ERCOT market conditions — bridging grid ops experience with BESS commercial logic.
- **Smart Security Light:** Designed ML pipeline and server subsystem using Python and CV2 to classify and track moving objects from infrared camera feed in real time.

Certifications & Professional Development

- Amazon AWS Developer Certification (07/2021)
- Texas A&M Students Serving Scouting — President, led organization through record year of growth (2021 – 2022)